

Community detection in geometric graphs: Recent advances until 2026

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Résumé TODO : introduction UGCM, presenter la litterature

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1 Introduction

Community detection consists in finding clusters in a graph where nodes share more connections with nodes of the same cluster than with nodes of other clusters. This broad topic is detailed in [Fortunato, 2010]. In this report, we focus on random geometric graphs (RGG). Adding spatial information allows us to consider local information based on real or abstract proximity. In social networks, two individuals are more likely to be friends if they are geographically close, or if they share similar interests, age, etc. More concrete examples of RGG can be found in wireless networks [Gomez et al., 2015], in biology [Higham et al., 2008], and also in community detection problems [Sankararaman et al., 2020] word embedding, physic,

The main reference model for community detection is the Stochastic Block Model (SBM) [Holland et al., 1983]. Each node carries a hidden community label and edge probabilities depend on whether two nodes belong to the same community or not. This model comes from the Erdős-Rényi graph, where each node connects to others with the same probability [Erdős et al., 1960]. This model is now well understood, see [Abbe, 2018]. Nonetheless, it has some drawbacks, since it is unrealistic to think that every node can connect to every other one with the same probability scale. Around the same time as Erdős-Rényi, the RGG was introduced by [Gilbert, 1961]. Adding geometric information allows us to mimic some emergent behaviours of real networks, such as sparsity [Del Genio et al., 2011], the "small-world" property [Watts and Strogatz, 1998] or transitivity, namely the idea that a friend of a friend is more likely to be a friend.

RGG and its properties are well studied, see [Penrose, 2003] and more recently [Duchemin and De Castro, 2023]. When we add community structure, we are in a different setting. We observe the graph, and sometimes also the node locations, while the community labels remain hidden. The natural question is then whether the latent structure can be detected, and if so, how accurately the labels can be recovered. We are interested in several levels of inference. First,

distinguishability asks whether the observed graph comes from a geometric model with communities or from a null model without community structure. This is a testing problem. It aims to distinguish the presence of a community structure, not to recover the labels. Then, if recovery is possible, one can ask how accurate it can be.

- *Weak/partial recovery* : detecting communities better than random guessing
- *Almost exact recovery* : the fraction of misclassified vertices vanishes as the number of nodes grows.
- *Exact recovery* : all community labels are correctly recovered, up to permutation.

Thus, in this report, community detection is not only understood as producing a convenient partition of the observed graph, but rather as recovering latent community labels under a probabilistic geometric model.

Previously, we emphasized properties that RGG share with real networks due to the geometric information. However, one can also obtain transitivity, and surely more, with attribute-based models [Li et al., 2018], hypergraphs [Ruggeri et al., 2023] and even geometric hypergraphs [Kovács et al., 2025]. We will not consider these models here, in order to keep the focus on the geometric aspect. Moreover, among the problems mentioned above, one is *Distinguishability* which should not be confused with the *Detection Problem* that we do not study here. The latter usually asks for conditions on the parameters defining the RGG under which the model is distinguishable from an Erdős-Rényi graph. Hence, this is not directly a community question. See, [Bubeck et al., 2016] [Liu et al., 2022] [Mao et al., 2026]. Finally, at the beginning of the literature, some works were interested in recovering the latent position of nodes in the geometric space, and considered this as a community detection problem, [Valdivia and De Castro, 2019] [Eldan et al., 2022]. This is a different problem from the one we are interested in, and it's in some sense harder, because one wants to recover the position (a continuous label) of each node rather than a community (a discrete label).
TODO : parler des gaussian-mixture model + trouver ref; voir commentaire latex !

2 Problem setting and inference tasks

Formally, we observe a graph $G := (V, E)$, where V is the vertex set and E is the set of edges between vertices. Throughout the report, n denotes the asymptotic scaling parameter. Depending on the model, the observed number of vertices may be either deterministic or random, but is of order n . We consider a simple undirected graph **Une remarque là dessus? Pourquoi non-orienté est classique comme hypothèse?**, without self-loops. The vertices are partitioned into k communities $C_1, \dots, C_k$, so $V = \bigsqcup_{i=1}^k C_i$. We denote by $\sigma^* : V \rightarrow [k]$ the ground-truth community label function, defined by $\sigma^*(v) = i$ whenever $v \in C_i$. The community structure is hidden, and our goal is to recover it from the observed data.

An estimator is therefore a function of the observations. Depending on the model, the observations may consist only of the graph G , or of the graph together with the node locations $X = (X_v)_{v \in V}$. Accordingly, we write the estimator as $\hat{\sigma} = \hat{\sigma}(G)$ or $\hat{\sigma} = \hat{\sigma}(G, X)$, and we view $\hat{\sigma} : V \rightarrow [k]$ as an estimated labeling. In general, the labels are only identifiable up to a relabeling of the communities. Indeed, permuting the names of the communities does not change the underlying partition of the vertex set. For this reason, the quality of an estimator is measured up to permutation. Denoting by $\mathfrak{S}_k$ the symmetric group on $[k]$, we define

$$A_n(\hat{\sigma}, \sigma^*) := \frac{1}{|V|} \max_{\pi \in \mathfrak{S}_k} \sum_{u \in V} \mathbf{1}\{\hat{\sigma}(u) = \pi(\sigma^*(u))\}$$

Thus, $A_n(\hat{\sigma}, \sigma^*)$ measures the fraction of correctly classified vertices after the best possible relabeling of the communities. We now give more formal definitions of the problems introduced in Section 1.

Distinguishability. Following [Sankararaman and Baccelli, 2018], distinguishability is a hypothesis testing problem. Let $(\mathbb{P}_n)_{n \geq 1}$ be the laws of a geometric random graph model with community structure and let $(\mathbb{Q}_n)_{n \geq 1}$ be the laws of a null geometric model without communities, both defined on the same observation space. Let Ω_n be the observation space. A test is a measurable function $\varphi_n : \Omega_n \rightarrow \{0, 1\}$ where $\varphi_n = 1$ means that we decide in favor of the community model. We say that $(\mathbb{P}_n)$ and $(\mathbb{Q}_n)$ are distinguishable if there exist tests $(\varphi_n)_{n \geq 1}$ and a constant $\gamma > 0$ such that, under the uniform prior on $\{\mathbb{P}_n, \mathbb{Q}_n\}$,

$$\mathbb{P}(\varphi_n \text{ makes the correct decision}) = \frac{1}{2} \mathbb{P}_n(\varphi_n = 1) + \frac{1}{2} \mathbb{Q}_n(\varphi_n = 0) \geq \frac{1}{2} + \gamma - o_n(1)$$

Weak/Partial recovery. We say that weak/partial recovery is achievable if the estimator performs better than random guessing, that's to say in the symmetric case, if there exists a sequence of estimators $(\hat{\sigma}_n)_{n \geq 1}$ and a constant $\alpha > 1/k$ such that,

$$\mathbb{P}(A_n(\hat{\sigma}_n, \sigma^*) \geq \alpha) \xrightarrow{n \rightarrow \infty} 1$$

Almost exact recovery. We say that almost exact recovery is achievable if there exists a sequence of estimators $(\hat{\sigma}_n)_{n \geq 1}$ such that for every $\varepsilon > 0$,

$$\mathbb{P}(A_n(\hat{\sigma}_n, \sigma^*) \geq 1 - \varepsilon) \xrightarrow{n \rightarrow \infty} 1$$

Exact recovery. We say that exact recovery is achievable if there exists a sequence of estimators $(\hat{\sigma}_n)_{n \geq 1}$ such that,

$$\mathbb{P}(A_n(\hat{\sigma}_n, \sigma^*) = 1) \xrightarrow{n \rightarrow \infty} 1$$

Most works in the literature focus on the case $k = 2$. Although several ideas extend beyond this setting, this is also the framework that we adopt in the

following, except when specified otherwise. In that case, it is convenient to encode the labels by $\{-1, +1\}$, instead of $[2] = \{1, 2\}$. Then “up to permutation” reduces to a global sign flip.

Each vertex also belongs to a space $\mathcal{X}$. In the following, we specify each time whether this space is latent or observed. For theoretical analysis, $\mathcal{X}$ is often chosen for mathematical convenience. The Euclidean space $\mathbb{R}^d$ is a natural reference because of its symmetry, homogeneity, isotropy, and direct geometric interpretation. See [Abbe et al., 2021], **Citer ceux qui sont euclidiens**. We also encounter the torus, mainly to avoid boundary effects and to focus on the core ideas of the proofs, **Todo citer**. Lastly, for latent models, the high-dimensional unit sphere $\mathbb{S}^d$ is a standard reference space to model similarity, and it also provides convenient spectral properties [Castro et al., 2020]. A summary of theoretical works on $\mathbb{S}^d$ is given in [Duchemin and De Castro, 2023].

In Section 3 we present models used throughout the literature. Hence, we suggest a model to unify them while remaining close to the notation used in the papers. We call it the *Unified Geometric Community Model* (UGCM) and it is defined as,

$$\text{UGCM}(\mathcal{X}, \rho, W_n, \mu_n, Z, \pi, \mathcal{Y}, K_n)$$

where $\mathcal{X}$ is the ambient space and ρ is its metric. $W_n \subset \mathcal{X}$ is the window where nodes are sampled at the scale n . μ_n governs the spatial sampling mechanism. Usually, the vertices are sampled either as n i.i.d. points in W_n with law μ_n or as a Poisson Point Process (PPP) on W_n with intensity measure μ_n . We denote by Z the label set, and by $\pi = (\pi_i)_{i \in Z}$ the distribution of the labels. For all $u \in V$ and independently, $\sigma^*(u) \sim \pi$. $\mathcal{Y}$ the space of pairwise observation. Let $\mathcal{P}(\mathcal{Y})$ denote the set of probability distributions on $\mathcal{Y}$, we consider the kernel observation,

$$K_n : \mathcal{X} \times \mathcal{X} \times Z \times Z \rightarrow \mathcal{P}(\mathcal{Y})$$

and for each unordered pair of vertices u, v , we observe,

$$Y_{uv} \mid (X, \sigma^*) \sim K_n(X_u, X_v, \sigma^*(u), \sigma^*(v))$$

The kernel is the core of the model. We note that Soft-RGG introduced by [Penrose, 2016] and developed in [Duchemin and De Castro, 2023] can be seen as a special case of UGCM where there exists a function $f_n : \mathbb{R}^+ \rightarrow [0, 1]$ such that for every $x, y \in \mathcal{X}$ and $z, z' \in Z$, $K_n(x, y, z, z') = \text{Ber}(f_n(\rho(x, y)))$. Moreover, a particular case of Soft-RGG are the usual random geometric graph, i.e. the hard-threshold special case, when f_n is an indicator function, i.e. there exists $r_n > 0$ such that $K_n(x, y, z, z') = \delta_{\mathbf{1}_{\{\rho(x, y) \leq r_n\}}}$. Then, this framework also accommodates weighted graphs, since the observation space $\mathcal{Y}$ is not necessarily binary.

As we will see, several phase-transition phenomena depend on the connectivity scale of the graph, which is usually measured through the average degree. Let D_n denote the degree of a vertex chosen uniformly at random. We can distinguish three regimes of connectivity, depending on the growth of $\mathbb{E}(D_n)$ as

$n \rightarrow \infty$. The most realistic but also the most delicate regime is the *sparse regime*. Each vertex only carries a finite amount of local information, the expected degree remains bounded,

$$\mathbb{E}(D_n) = \Theta(1)$$

In the *logarithmic regime*, the graph becomes sufficiently connected for exact recovery to become meaningful,

$$\mathbb{E}(D_n) = \Theta(\log n)$$

Finally, the *dense regime* corresponds to the easiest setting, the graph carries a lot of information,

$$\mathbb{E}(D_n) = \Theta(n)$$

3 Geometric community models and recovery thresholds

Faire le rapport dans l'ordre chronologique je pense

- [Galhotra et al., 2018] (Geometric Block Model, GBM)
- [Sankararaman and Baccelli, 2018] & [Abbe et al., 2021] (Geometric SBM)
- [Chien et al., 2020]
- [Galhotra et al., 2020] (GBM)
- [Galhotra et al., 2023] (GBM)
- [Avrachenkov et al., 2024] (Geometric Kernel Block Model, GKBM)
- [Gaudio et al., 2024b] (Geometric SBM)
- [Gaudio et al., 2024a] (Geometric Hidden Community Model, GHCM)
- [Gaudio and Guan, 2025] (GHCM)
- [Gaudio and Jin, 2026] (GHCM)

3.1 Geometric Block Model (GBM)

[Galhotra et al., 2018] [Galhotra et al., 2020] [Galhotra et al., 2023]
[Chien et al., 2020]

3.2 Geometric Stochastic Block Model (GSBM)

[Sankararaman and Baccelli, 2018] [Abbe et al., 2021]

3.3 (Geometric Kernel Block Model, GKBM)

[Avrachenkov et al., 2024] [Allem et al., 2025]

3.4 Geometric Hidden Community Model (GHCM)

[Gaudio et al., 2024b] [Gaudio et al., 2024a] [Gaudio and Guan, 2025] [Gaudio and Jin, 2026]

4 Spectral

(A confirmer) [Chen et al., 2016]
 (papier SBM mais théorie intéressante) [Bordenave et al., 2015]
 Cadre spectral intéressant pour certains espaces [Méliot, 2019]
 Méthode spectrale + Robustesse (la perturbation est géométrique) [Péché and Perchet, 2020]
 Méthode spectrale sur Soft Geometric Block Model (SGBM) (Generalisation de Geometric SBM) [Avrachenkov et al., 2021]

5 Ouvertures

- Considérer des modèles hyperboliques, pas seulement labels & distance-dependant, car ceux-ci captent mieux les propriétés habituelles des réseaux
 - Clustering [Krioukov, 2016]
 - Degree inhomogène [Barabási and Albert, 1999]
 - "Communities with specific mixing patterns" (mesoscale) [Toivonen et al., 2006]
- Communautés apparaissent dans un espace hyperbolique latent [Bruno et al., 2019] Réponse 6ans plus tard : pas toujours! + définition d'un modele pour "régler" ça [Guarino et al., 2025]
 [Li et al., 2018]

Bibliographie

- Emmanuel Abbe. Community detection and stochastic block models : recent developments. *Journal of Machine Learning Research*, 18(177) :1–86, 2018.
- Emmanuel Abbe, Francois Baccelli, and Abishek Sankararaman. Community detection on euclidean random graphs. *Information and Inference : A Journal of the IMA*, 10(1) :109–160, 2021.
- Luiz Emilio Allem, Konstantin Avrachenkov, Carlos Hoppen, Hariprasad Manjunath, and Lucas Siviero Sibemberg. Multi-community spectral clustering for geometric graphs, 2025. URL <https://arxiv.org/abs/2508.00893>.
- Konstantin Avrachenkov, Andrei Bobu, and Maximilien Drevet. Higher-order spectral clustering for geometric graphs. *Journal of Fourier Analysis and Applications*, 27(2) :22, 2021.
- Konstantin Avrachenkov, BR Kumar, and Lasse Leskelä. Community detection on block models with geometric kernels. *arXiv preprint arXiv :2403.02802*, 2024.
- Albert-László Barabási and Réka Albert. Emergence of scaling in random networks. *Science*, 286(5439) :509–512, October 1999. ISSN 1095-9203. <https://doi.org/10.1126/science.286.5439.509>. URL <http://dx.doi.org/10.1126/science.286.5439.509>.
- Charles Bordenave, Marc Lelarge, and Laurent Massoulié. Non-backtracking spectrum of random graphs : community detection and non-regular ramanujan graphs. In *2015 IEEE 56th Annual Symposium on Foundations of Computer Science*, pages 1347–1357. IEEE, 2015.
- Matteo Bruno, Sandro Ferreira Sousa, Furkan Gursoy, Matteo Serafino, Francesca V Vianello, Ana Vranić, and Marián Boguñá. Community detection in the hyperbolic space. *arXiv preprint arXiv :1906.09082*, 2019.
- Sébastien Bubeck, Jian Ding, Ronen Eldan, and Miklós Z Rácz. Testing for high-dimensional geometry in random graphs. *Random Structures & Algorithms*, 49(3) :503–532, 2016.
- Yohann De Castro, Claire Lacour, and Thanh Mai Pham Ngoc. Adaptive estimation of nonparametric geometric graphs, 2020. URL <https://arxiv.org/abs/1708.02107>.
- Yuxin Chen, Govinda Kamath, Changho Suh, and David Tse. Community recovery in graphs with locality. In *International conference on machine learning*, pages 689–698. PMLR, 2016.
- Eli Chien, Antonia Tulino, and Jaime Llorca. Active learning in the geometric block model. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pages 3641–3648, 2020.
- Charo I. Del Genio, Thilo Gross, and Kevin E. Bassler. All scale-free networks are sparse. *Physical Review Letters*, 107(17), October 2011. ISSN 1079-7114. <https://doi.org/10.1103/physrevlett.107.178701>. URL <http://dx.doi.org/10.1103/PhysRevLett.107.178701>.

- Quentin Duchemin and Yohann De Castro. Random geometric graph : Some recent developments and perspectives. *High Dimensional Probability IX : The Ethereal Volume*, pages 347–392, 2023.
- Ronen Eldan, Dan Mikulincer, and Hester Pieters. Community detection and percolation of information in a geometric setting. *Combinatorics, Probability and Computing*, 31(6) :1048–1069, 2022.
- Paul Erdős, Alfréd Rényi, et al. On the evolution of random graphs. *Publications of the*, 1960.
- Santo Fortunato. Community detection in graphs. *Physics reports*, 486(3-5) : 75–174, 2010.
- Sainyam Galhotra, Arya Mazumdar, Soumyabrata Pal, and Barna Saha. The geometric block model. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, 2018.
- Sainyam Galhotra, Arya Mazumdar, Soumyabrata Pal, and Barna Saha. Connectivity in random annulus graphs and the geometric block model, 2020. URL <https://arxiv.org/abs/1804.05013>.
- Sainyam Galhotra, Arya Mazumdar, Soumyabrata Pal, and Barna Saha. Community recovery in the geometric block model, 2023. URL <https://arxiv.org/abs/2206.11303>.
- Julia Gaudio and Charlie K. Guan. Sharp exact recovery threshold for two-community euclidean random graphs, 2025. URL <https://arxiv.org/abs/2501.14830>.
- Julia Gaudio and Andrew Jin. Exact recovery in the geometric hidden community model. *arXiv preprint arXiv :2601.17591*, 2026.
- Julia Gaudio, Charlie Guan, Xiaochun Niu, and Ermin Wei. Exact label recovery in euclidean random graphs. *arXiv preprint arXiv :2407.11163*, 2024a.
- Julia Gaudio, Xiaochun Niu, and Ermin Wei. Exact community recovery in the geometric sbm, 2024b. URL <https://arxiv.org/abs/2307.11196>.
- Edward N Gilbert. Random plane networks. *Journal of the society for industrial and applied mathematics*, 9(4) :533–543, 1961.
- Jean-Sébastien Gomez, Aurélien Vasseur, Anaïs Vergne, Philippe Martins, Laurent Decreusefond, and Wei Chen. A case study on regularity in cellular network deployment. *IEEE wireless communications letters*, 4(4) :421–424, 2015.
- Stefano Guarino, Enrico Mastrostefano, and Davide Torre. Random hyperbolic graphs with arbitrary mesoscale structures. *Physical Review E*, 112(5) :054310, 2025.
- Desmond J Higham, Marija Rašajski, and Nataša Pržulj. Fitting a geometric graph to a protein–protein interaction network. *Bioinformatics*, 24(8) :1093–1099, 2008.
- Paul W Holland, Kathryn Blackmond Laskey, and Samuel Leinhardt. Stochastic blockmodels : First steps. *Social networks*, 5(2) :109–137, 1983.
- Bianka Kovács, Barnabás Benedek, and Gergely Palla. Community detection in hypergraphs through hyperedge percolation. *Scientific Reports*, 15(1), October 2025. ISSN 2045-2322. <https://doi.org/10.1038/s41598-025-19974-9>. URL <http://dx.doi.org/10.1038/s41598-025-19974-9>.

- Dmitri Krioukov. Clustering implies geometry in networks. *Physical Review Letters*, 116(20), May 2016. ISSN 1079-7114. <https://doi.org/10.1103/physrevlett.116.208302>. URL <http://dx.doi.org/10.1103/PhysRevLett.116.208302>.
- Ye Li, Chaofeng Sha, Xin Huang, and Yanchun Zhang. Community detection in attributed graphs : An embedding approach. *Proceedings of the AAAI Conference on Artificial Intelligence*, 32(1), Apr. 2018.
- Siqi Liu, Sidhanth Mohanty, Tselil Schramm, and Elizabeth Yang. Testing thresholds for high-dimensional sparse random geometric graphs. In *Proceedings of the 54th Annual ACM SIGACT Symposium on Theory of Computing*, pages 672–677, 2022.
- Cheng Mao, Yihong Wu, and Jiaming Xu. Random geometric graphs with smooth kernels : sharp detection threshold and a spectral conjecture, 2026. URL <https://arxiv.org/abs/2602.14998>.
- Pierre-Loïc Méliot. Asymptotic representation theory and the spectrum of a random geometric graph on a compact lie group. 2019.
- Sandrine Péché and Vianney Perchet. Robustness of community detection to random geometric perturbations. *Advances in Neural Information Processing Systems*, 33 :17827–17837, 2020.
- Mathew Penrose. *Random geometric graphs*, volume 5. OUP Oxford, 2003.
- Mathew D. Penrose. Connectivity of soft random geometric graphs. *The Annals of Applied Probability*, 26(2), April 2016. ISSN 1050-5164. <https://doi.org/10.1214/15-aap1110>. URL <http://dx.doi.org/10.1214/15-AAP1110>.
- Nicolò Ruggeri, Martina Contisciani, Federico Battiston, and Caterina De Bacco. Community detection in large hypergraphs. *Science Advances*, 9(28), July 2023. ISSN 2375-2548. <https://doi.org/10.1126/sciadv.adg9159>. URL <http://dx.doi.org/10.1126/sciadv.adg9159>.
- Abishek Sankararaman and François Baccelli. Community detection on euclidean random graphs. In *Proceedings of the Twenty-Ninth Annual ACM-SIAM Symposium on Discrete Algorithms*, pages 2181–2200. SIAM, 2018.
- Abishek Sankararaman, Haris Vikalo, and François Baccelli. Comhapdet : a spatial community detection algorithm for haplotype assembly. *BMC genomics*, 21(Suppl 9) :586, 2020.
- Riitta Toivonen, Jukka-Pekka Onnela, Jari Saramäki, Jörkki Hyvönen, and Kimmo Kaski. A model for social networks. *Physica A : Statistical Mechanics and its Applications*, 371(2) :851–860, November 2006. ISSN 0378-4371. <https://doi.org/10.1016/j.physa.2006.03.050>. URL <http://dx.doi.org/10.1016/j.physa.2006.03.050>.
- Ernesto Araya Valdivia and Yohann De Castro. Latent distance estimation for random geometric graphs. In *NeurIPS*, pages 8721–8731, 2019.
- Duncan J Watts and Steven H Strogatz. Collective dynamics of 'small-world' networks. *nature*, 393(6684) :440–442, 1998.